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Human Mobility Datasets in the Complex Metro System of Shanghai

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The growing role of metro systems in urban mobility calls for high-quality metro transit datasets. Derived from over 700 million smart card records, an open-sourced, city-scale metro flow dataset was constructed, covering the period of May-August 2017 and 302 metro stations in Shanghai, China. The in-out flow counts of each station and OD flow between stations were offered at a 10-minute temporal resolution. By leveraging the mobility patterns of each passenger, metro flows were categorized into commuting flows, home-based-other flows, and none-home-based flows, providing a more comprehensive perspective towards urban mobility dynamics. Supplemental metadata, including station attributes, network topology, and meteorological records further support potential applications. This city-scale metro flow dataset could be utilized in advancing research in transportation modeling, spatio-temporal data mining, and urban mobility analysis.

Background & Summary

Human mobility, the spatial-temporal movement patterns of people, provides fundamental insights for urban science and city governance[1]. It supports a wide range of applications including transportation optimization[2,3], city planning[4,5], epidemic modeling[6,7], and environmental impact assessment[8–10]. Among various modes of urban transportation, subway systems stand out for their high travel efficiency, large passenger capacity, and low environmental impact. By 2020, more than 190 metro systems had been established globally. Around 20%–50% of commuters travel by subway every day in major metropolitan areas such as Paris, London, and Tokyo[11].

The increasingly important role of subway system in urban transportation has sparked numerous studies leveraging metro transit data, which could be broadly categorized into passenger behavior analysis, operation optimization, and policy applications[12]. Utilizing smart card data from Seoul, researchers proposed an agent-based transportation simulator to analyze waiting time and transfer volumes across different metro stations[13]. Passengers' trip chains could be identified and clustered based on spatiotemporal similarities[14]. Mobility patterns of passengers extracted from the transit data offer valuable guidance for optimizing system operations, such as minimizing total waiting time[15,16] and reducing operation cost[17,18]. At a broader scale, metro transit data sheds light on city spatial structure, urbanization process, and local policy decisions[19,20]. Furthermore, the prevalence of artificial intelligence and big data technologies facilitated the development of data-driven metro flow prediction models[21–23], which rely heavily on metro transit data.

Despite the growing body of research leveraging metro transit data, high-quality datasets accessible to the broader research community remain scarce. The Metropolitan Transportation Authority (MTA), operator of public transportation in the New York City metropolitan area, has released a series of subway ridership datasets spanning five years and covering more than 400 metro stations. The MTA Subway Hourly Ridership Dataset (https://data.ny.gov/Transportation/MTA-Subway-Hourly-Ridership-2020-2024/wujg-7c2s/about_data) records the number of riders entering each station on an hourly basis starting from 2020, with the most recent update on March 5, 2025. The MTA Subway Origin-Destination Ridership Dataset (https://data.ny.gov/browse?tags=origin-destination&sortBy=relevance&pageSize=20) estimates the number of riders that traveled between specific origin-destination pairs for each hour of day and day of the week, averaged over a calendar month. Though monthly-averaged OD flow provided more detailed mobility patterns than ridership, its accuracy is

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limited because trip destinations are inferred based on users' subsequent entering swipes. The MTA Subway Turnstile Usage Dataset (https://data.ny.gov/browse?q=MTA+Subway+Turnstile+Usage+Data&sortBy=relevance&page=1&pageSize=20) further refines the hourly ridership dataset by providing the entry and exit counts for each turnstile. Beyond the MTA Subway Dataset, few datasets rival these in terms of scale, standardization, and maintenance frequency. SHMetro and HZMetro (https://lingboliu.com/Dataset.html) are two datasets first proposed by Linbo Liu[23] and become benchmarks for metro flow prediction task[24,25]. SHMetro records the inflow and outflow of 288 metro stations in Shanghai from July to September 2016, while HZMetro captures similar data for 80 metro stations in Hangzhou from January 1 to January 25, 2019. Both datasets feature a 15-minute temporal resolution. However, they are primarily designed for time-series forecasting tasks, and the absence of station names or precise locations creates discrepancies with real-world facilities.

In light of the dataset deficit, we compiled a large-scale, high-resolution, information-rich metro flow dataset[26] to capture the dynamic mobility pattern in Shanghai's subway system. As a global financial hub and a major center for commerce, technology, and culture, Shanghai is located on the eastern coast of China, spanning 6,340.5 square kilometers with a population of 24.87 million. In this work, we extracted over 736.8 million metro trip records from the smart card data collected by the Automatic Fare Collection (AFC) system, covering 302 metro stations, extending from May 1st to August 31st, 2017. By leveraging personal trajectory data across 19.69 million users, all trips were categorized into commuting trips, home-based-others trips, and none-home-based trips. To enhance privacy protection while effectively illustrating mobility patterns, we aggregated trips into station-level inflow/outflow counts and origin-destination (OD) pair flows, all at a 10-minute temporal resolution.

We summarized the key contributions of this work as follows:

- We compiled and released a large-scale metro flow dataset capturing the mobility patterns of Shanghai's subway system across 302 metro stations over four months. Additionally, we collected station attributes, network topology, and weather data in our data repository.
- By analyzing individuals' trajectories, we distinguished commuting trips and home-based-others trips from none-home-based trips, offering more comprehensive insights and detailed data for related studies.
- We conducted systematic technical validations, including integrity checks, correlation analysis, and comparisons with a reference dataset, to verify the dataset's robustness and accuracy.

Methods

In this section, we systematically described the data organization process and the specific techniques employed in its implementation. We begin with a general overview, followed by a detailed discussion about data acquisition, data cleaning, and trip labeling.

As illustrated in Fig. 1, the data organization workflow begins with transaction records generated by the automatic fare collection (AFC) system. A subway trip was completed when a passenger tapped their card at both entry and exit points. The original smart card data spans from May 1st to August 31st, 2017, covering a total of 123 days. Given that the typical operation hours of the Shanghai subway system are from 6 A.M. to 11 P.M., we segmented each day into 102 discrete 10-minute intervals. During each interval, entry counts, exit counts, and origin-destination flow were aggregated across 302 stations, with OD flows aligned by entry time. Figure 2(a) demonstrates the system's total ridership on workdays and non-workdays. A distinctive pattern on workdays is the morning rush hours (7:00–9:00 A.M.) and evening rush hours (5:00–7:00 P.M.).

By further analyzing the commuting patterns within daily mobility behaviors, we managed to identify the residence station and workplace station for over 67% of smart card users. With this information, we labeled all the trips as commuting trips, home-based-others trips, and none-home-based trips. Figure 2(b) reveals the probability distribution of travel distances for all kinds of trips. The curve peaks at $d = 10$, indicating that the typical distance of a subway trip is around 10 km. Beyond this peak, the probability drops dramatically following an exponential distribution (green line), which is consistent with the truncated power-law tail observed in classical human mobility studies[27]. The power law fails to fit trips shorter than 10 km due to the prevalence of alternative transportation modes such as buses and bicycles. Interestingly, comparing the proportion of short-distance trips in all kinds of trips, we observed that the none-home-based trips rank first, followed by home-based-others trips and commuting trips, while the pattern is the complete opposite for long-distance trips. Commuting trips exhibit the heaviest tail in the distribution.

Based on 736.8 million extracted trips, the activity levels of 19.69 million users and the overall busyness of 302 metro stations are depicted by their total trip volumes aggregated over the entire period in Fig. 2(c). The travel frequencies of users follow an exponential distribution. The majority of metro stations experienced trip volumes less than 4 million over a total of 123 days, with a few exceptions such as stations located at transportation hubs or downtown areas.

Figure 3 demonstrates our station-level high-resolution metro flow dataset, comparing the in-out passenger flows of five major stations using data from May 1st (a non-workday) and May 2nd (a workday). People's Square, along with Xujiahui and Lujiazui, is located in the downtown area of Shanghai, combining the functions of a commercial district and a tourist attraction. On the workday, the commuting outflow counts of these three stations peaked at 8 A.M. while inflow counts peaked at 6 P.M., reflecting a clear workplace-driven pattern. In contrast, Xinzhuang Station exhibits distinct morning inflow and evening outflow patterns, revealing its role as a commuter hub on workdays. Consequently, its passenger volumes on non-workdays are significantly lower than those at the other stations. The none-home-based trips take the lead in the flow counts of Hongqiao Railway Stations, which indicates a large proportion of users here are transient populations or intercity travelers. As a key railway hub, Hongqiao Railway Station witnessed significantly lower trip volume on workdays than on non-workdays.

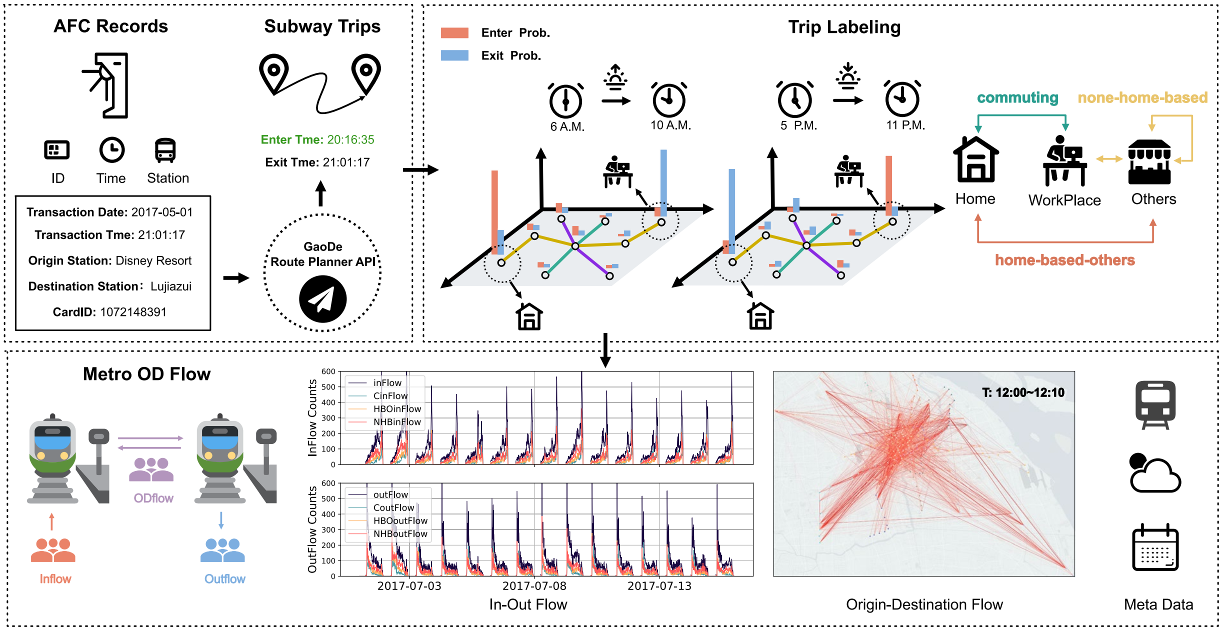

Fig. 1 Schematic overview of the study. This study was initiated using smart card transaction data collected through the automatic fare collection (AFC) system. A complete subway trip was determined after inferring the entry time via Gaode API. For each user, the residence and workplace stations were inferred from personal trajectories during commuting time. Then we further labeled trips as commuting trips (C), home-based-others (HBO) trips, and none-home-based (NHB) trips. Finally, these labelled trips were aggregated into inflow, outflow counts, and OD flow counts in a temporal resolution of 10 minutes, with metadata attached.

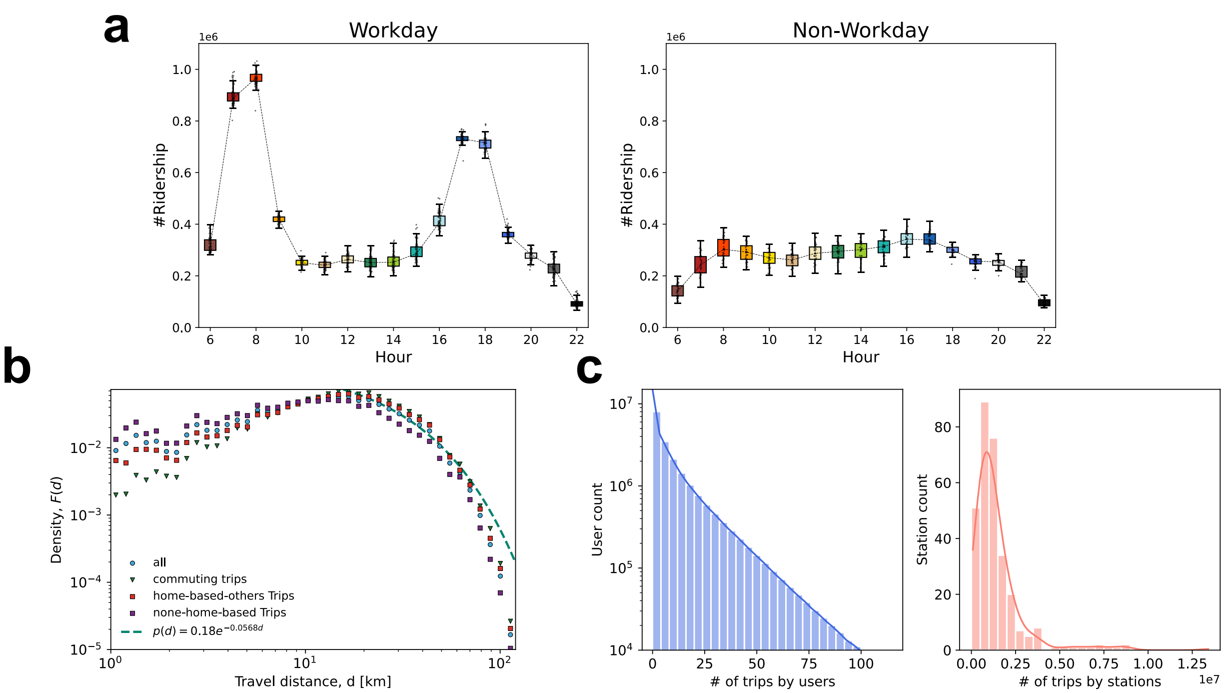

Fig. 2 (**a**) The total ridership of the subway system aggregated by hour for both workdays and non-workdays. Ridership is calculated as the sum of inflow volumes across all metro stations. (**b**) Distribution of travel distances for different trip types. Four curves composed of various styles of points illustrate the probability distribution of travel distances for all trips, commuting trips, home-based-others trips, and none-home-based trips, respectively. The green dashed line represents the fitted distribution of all trips covering distances greater than 10 km. (**c**) Visualization of users' activity level and busyness of metro stations in terms of trip volume.

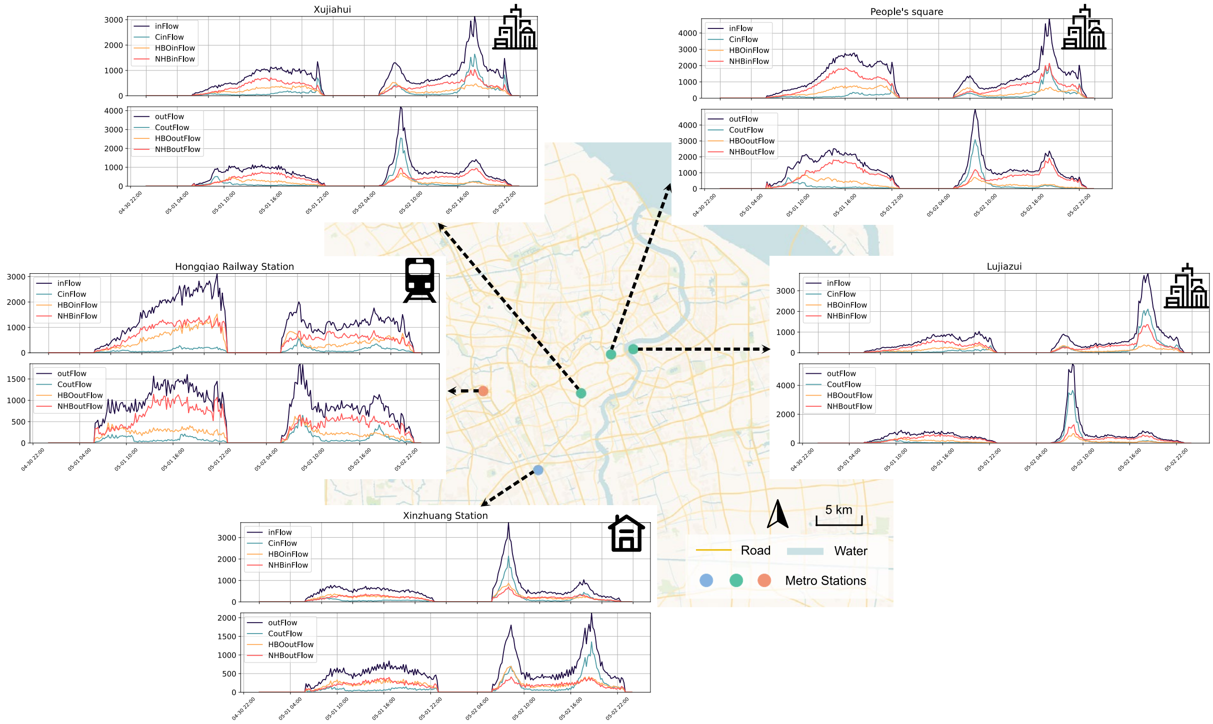

Fig. 3 A demonstration of the metro flow dataset. The charts visualized the in-out flow counts of five major metro stations on May 1st (a non-workday) and May 2nd (a workday) at a temporal resolution of 10 minutes. The multi-colored curves correspond to different flow types.

Data acquisition. The raw data, including station attribute data and smart card data, were obtained through the authors' affiliation with Shanghai Transportation Information Center, covering the transit records of 19.69 million users. To ensure privacy protection, all the user IDs had been anonymized before any analysis. Since our released dataset contains only aggregated flow data, no personally identifiable information is disclosed. We have received explicit approval from the Shanghai Transportation Information Center to publish the aggregated data under an open license.

Some other data involved in this study originates from third parties. The hourly weather data was obtained using the Historical Weather API of Open-meteo (https://open-meteo.com/en/docs/historical-weather-api). The estimated travel time and travel distance between two stations were acquired through Route Planning API 2.0 (Metro Mode) of Gaode Maps (https://lbs.amap.com/api/webservice/guide/api/newroute). We also handcrafted a workday calendar according to the 2017 Chinese Government Holiday Announcement (https://www.gov.cn/zhengce/content/2016-12/01/content_5141603.htm). All these data are under open license and integrated into our data repository except estimated travel time and travel distance, which could be accessed by subscribing to the service of Gaode.

Data cleaning. The original smart card data contains transaction records for all modes of public transportation. While preprocessing the data, we selected metro-specific entries and discarded records before 6:00 A.M. or after 11:00 P.M., which accounted for only a negligible portion of the dataset. A key challenge in data cleaning was the absence of the entry time for subway trips. Utilizing estimated travel time obtained via Gaode API, we inferred entry time based on the recorded entry station, exit station, and exit time.

To improve the reliability of the proposed dataset, anomalous smart card records were excluded by following criteria: 1) We eliminated smart card records that do not constitute a complete trip, specifically those lacking a corresponding tap-in (entry) or tap-out (exit) event; 2) We removed the metro trips with estimated durations shorter than 3 minutes or longer than 5 hours, as they are typically caused by turnstile faults, system errors, or highly atypical travel patterns, rather than normal transit behavior. After data cleaning, the integrity of metro flow dataset was validated in Fig. 4.

Trip labeling. To better understand the mobility patterns of metro transit dataset, we labeled trips as commuting (C) trips, home-based-others (HBO) trips and none-home-based (NHB) trips, defined as follows:

- ***commuting trips***: *Trips that occur between an individual's residence station and their workplace station.*
- ***home-based-others Trips***: *Trips that originate or terminate at an individual's residence station but are not commuting trips.*
- ***none-home-based Trips***: *Trips that neither originate nor terminate at an individual's residence station.*

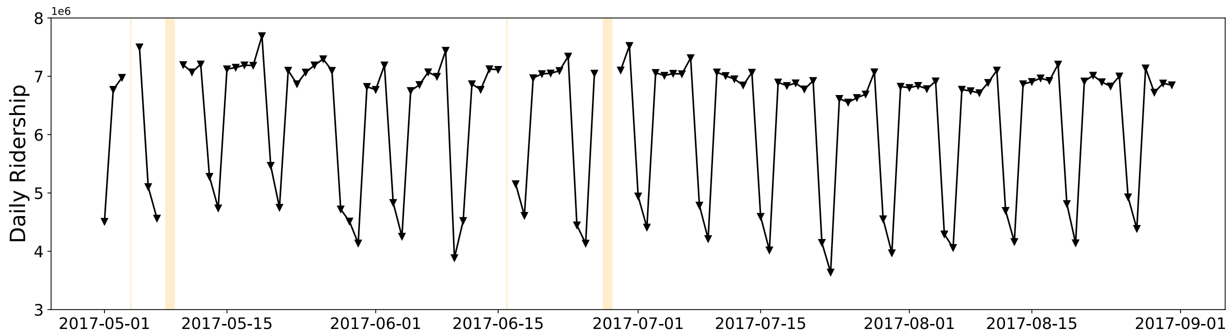

Fig. 4 Visualization of total daily ridership ranging from May 1st to August 31st. The accumulated flow counts on May 4th, May 8th, May 9th, June 16th, June 27th, and June 28th are significantly lower than usual, which are masked by orange blocks.

Date	Slot	StartTime	EndTime	Station	IF	OF	CIF	HIF	NIF	COF	HOF	NOF
20170501	0	060000	061000	112	29	32	9	15	5	11	13	8
20170501	0	060000	061000	113	142	102	40	51	51	30	41	31
20170501	0	060000	061000	114	66	35	17	19	30	12	10	13
…	…	…	…	…	…	…	…	…	…	…	…	…
20170831	12545	225000	230000	2057	2	36	0	0	2	13	9	14

Table 1. An example of Metro Station Inflow and Outflow Data.

According to the definition above, the key to trip labeling lies in the identification of the residence station and the workplace station for each user derived from their individual trajectory set. Specifically, a residence station pool *H* and workplace station pool *W* are maintained for each user. By traversing the trips during morning rush hours and evening rush hours, we assumed the entry stations and exit stations as potential residence stations and workplace stations accordingly, and put them into the pools with repetition. Finally, if a station appears most frequently in *H* (*W*) and accounts for more than 40%, it will be recognized as this user's residence station (workplace station). To summarize, stations frequently used as entry points in the morning and exit points in the evening are more likely to be identified as residence stations, and vice versa, as visualized in Fig. 1.

Data Records

Our Metro Dataset, available at figshare[26], consists of four components: Station Attributes Data, Inflow and Outflow Data, OD Flow Data, and Meta Data. The Station Attributes Data characterizes each metro station with ID, location (longitude, latitude), name, and neighboring stations which collectively represent the topological structure of the subway network. The Meta Data include a workday calendar to distinguish between workdays and non-workdays, and hourly weather information.

- Station Attributes Data (stationInfo.csv)
- Metro Station Inflow and Outflow Data (metroData_InOutFlow.csv)
- Metro Station Origin-Destination Flow Data (metroData_ODFlow.csv)
- Meta Data (shanghai_weatherHourly.csv, work_calendar.csv)

All the data above are stored in CSV (Comma-Separated Values) format with descriptive column headers, which is easy to read and operate across different devices and environments.

Examples of In-Out flow data and OD flow data are presented in Tables 1, 2, respectively. These two datasets provide continuous metro ridership data for 123 days, starting from 6 A.M. and finishing at 11 P.M. each day. Each row in Table 1 describes the total in-out flow as well as different types of flow at a single metro station within a specific 10-minute interval. The first four columns identify this interval: Date (concatenation of year, month, and day sequentially), Slot (a cumulative count of 10-minute intervals from '2017-05-01 06:00' UTC + 8), and StartTime & EndTime (concatenation of hour, minute, and second sequentially, in UTC + 8). The flow types include IF = total inflow, OF = total outflow, CIF = commuting inflow, HIF = home-based-others inflow, NIF = none-home-based inflow, COF = commuting outflow, HOF = home-based-others outflow, NOF = none-home-based outflow.

Similarly, as Table 2 illustrated, each row in OD Data illustrates the flow originating from O-Station and terminating at D-Station, where CFlow = commuting flow, HFlow = home-based-others flow, NFlow = none-home- based flow. The In-Out Flow Data and OD Flow Data represent two complementary perspectives of the same mobility dataset: the former is structured from the viewpoint of individual stations, while the latter provides a more detailed depiction of flows across the entire network.

Date	slot	StartTime	EndTime	O-Station	D-Station	Flow	CFlow	HFlow	NFlow
20170501	0	060000	061000	112	129	1	0	0	1
20170501	0	060000	061000	112	134	1	0	0	1
20170501	0	060000	061000	112	138	1	0	0	1
20170501	0	060000	061000	112	828	1	0	1	0
…	…	…	…	…	…	…	…	…	…
20170831	12545	225000	230000	2057	2036	1	0	0	1

Table 2. An example of Metro Station Origin-Destination Flow Data.

Technical Validation

Once the Metro Flow OD Dataset was constructed, a series of technical validation methods were implemented to verify the quality and reliability of the dataset: (1) Firstly, the integrity of the dataset was checked. (2) The correlation between metro flow data and weather data was analyzed. (3) We compared it with an open-source metro flow dataset from 2015 in Shanghai.

Data integrity. We provide continuous metro in-out flow data and OD flow data spanning from May 1st to August 31st in 2017, at a time interval of 10 minutes. However, the smart card data were partially missing on certain days, leading to incomplete flow counts. As illustrated in Fig. 4, the daily ridership on May 4th, May 8th, May 9th, June 16th, June 27th, and June 28th is significantly lower than that on other days, accounting for less than 2% of the total observation period.

Furthermore, by comparing each station's daily flow counts with its 123-day average, we observed that during these six days, all 302 stations exhibited ridership below 10% of their respective averages, indicating severe data insufficiency. No other days showed station-level data absence. Consequently, these six days were excluded from subsequent analyses.

Weather correlation analysis. Weather conditions significantly affect travel demand, which in turn influences metro ridership. We calculated the daily ridership of the subway system and analyzed its correlation with precipitation and temperature in Fig. 5(a). The 24-hour rainfall data were categorized into two classes based on a threshold of 5 millimeters, with rainfall exceeding this value indicating significant precipitation. The point-biserial correlation between ridership and precipitation is noticeably higher on non-workdays, along with a smaller p-value, revealing greater elasticity of travel demand to rainfall during non-workdays. The reduction of travel demand due to high temperature is evidenced by the negative pearson correlation coefficients between ridership and temperature on workdays and non-workdays. In summary, the intuitive correlation between metro flow and weather patterns extracted from data supports the reliability of the metro dataset.

Furthermore, an event study was conducted to analyze the impact of the rainstorm on June 10th, the second Saturday of June, 2017. We targeted the study on three specific stations: Station #1136 and Station #1628 are Disney Resort and Shanghai Wild Animal Park respectively, both of which are susceptible to rainstorms on weekends. Station #643 is located at the center of the rainstorm. As shown in Fig. 5(b), the inflow counts of these three stations dropped significantly compared to non-workday monthly average values during the extreme weather event.

Comparison to the reference data. Another open-source Shanghai metro transit dataset (https://lingboliu.com/Dataset.html) provides in-out flow counts of 288 metro stations from July to September 2015, at 15-minute intervals. Fortunately, with this dataset, we can identify the shared patterns between the reference dataset and our own, thereby validating the quality and reliability of this work.

However, the metro stations in the reference dataset were only identified by IDs, making it impossible to map them to specific physical locations. Thus, we focused on the comparative analysis of statistical regularities across the entire system. We divided a day into time bins of an hour for both workdays and non-workdays, resulting in a total of 34 time bins. First, the total flow counts of each time bin were aggregated across the two datasets and compared using scatter plots, as shown in the left panel of Fig. 6. Next, the variance of flow counts across all stations in each particular time bin was calculated and visualized in the right panel of Fig. 6. The Pearson Correlation Coefficients close to 1, along with low p-values, suggest that the patterns in our dataset closely match those of the reference dataset.

Usage Notes

The Station Attributes Data and Flow Data are offered in CSV format, which is easy to access and process. For storage efficiency, the OD Flow data was saved in a sparse matrix format (Table 2), where pairs of metro stations with no flow during specific intervals are excluded from the dataset. Despite this optimization, the OD data file remains large (12GB). The data on May 4th, May 8th, May 9th, June 16th, June 27th, and June 28th are incomplete due to deficiencies in the raw data. Researchers and analysts may apply data imputation techniques to address these gaps and utilize the dataset effectively.

Both the in-out flow data and OD flow data could be exploited in studies involving traffic optimization, data mining, human mobility, and urban science. For instance, station in-out flow data could be utilized to study cities' scaling rules[28] and community structure[29]. OD flow data contributes to the construction of a dynamic metro mobility network[30], unveiling important patterns which further facilitate ridership prediction[31] and urban planning[32].

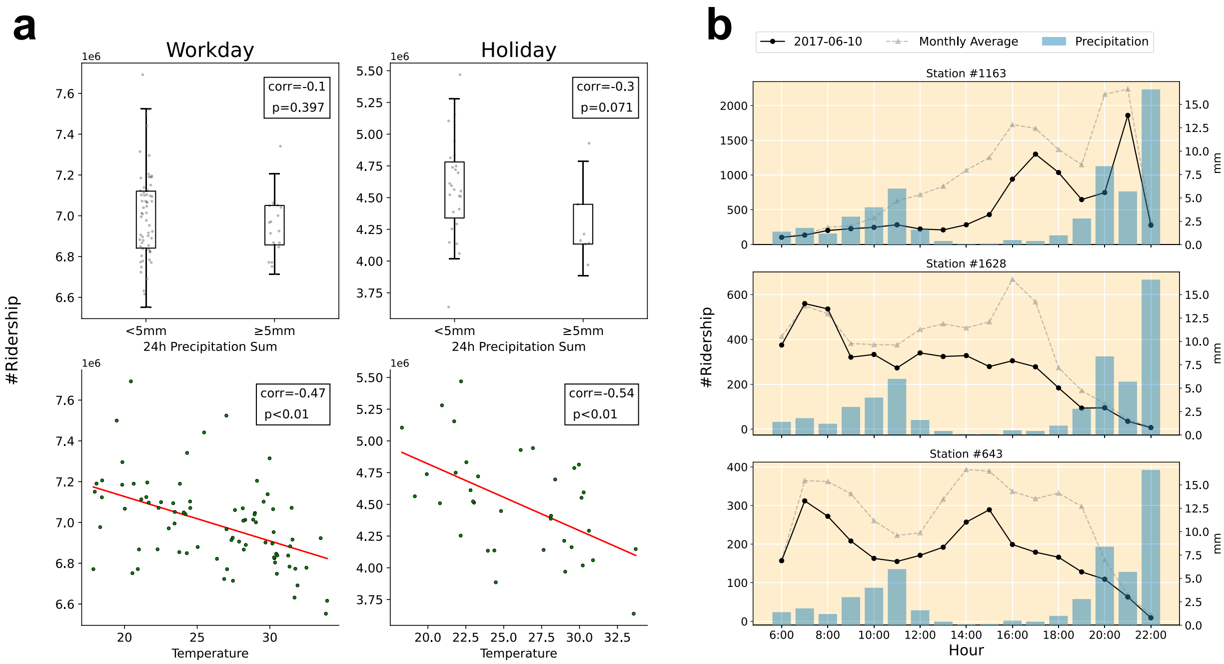

Fig. 5 (**a**) Correlation Analysis between ridership and weather: The first row of graphs presents an analysis of the relationship between total passenger flow and precipitation, while the second row shows the correlation between total passenger flow and temperature. Corr represents the point-biserial correlation and the pearson correlation coefficient, respectively, for rainfall and temperature. *p* stands for probability value. (**b**) Event Study: This analysis focuses on three metro stations on June 10th, during a heavy rainstorm in Shanghai. The two curves represent hourly ridership on the event day and the monthly average values on non-workdays. The bars indicate the recorded rainfall in millimeters.

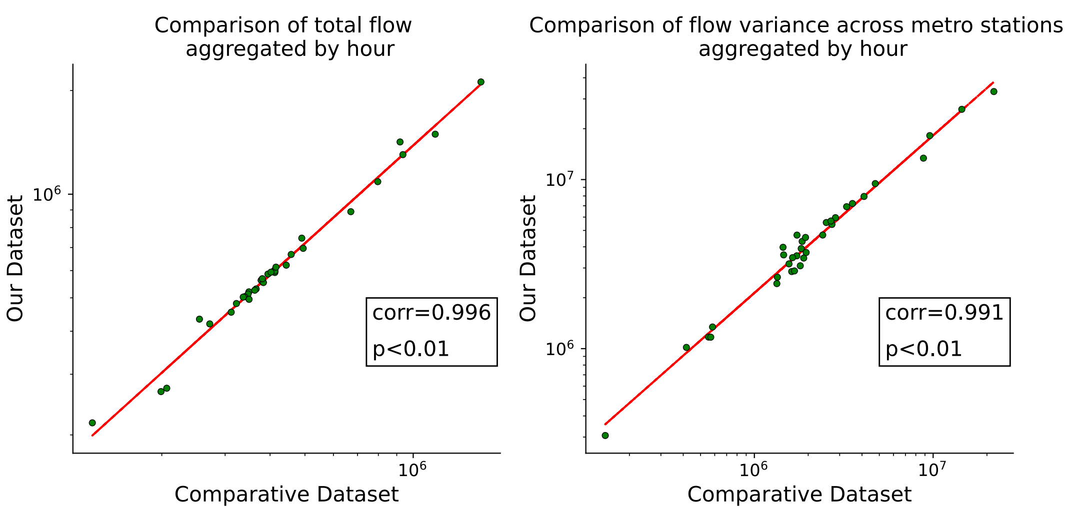

Fig. 6 Comparison of hourly flow counts and flow variance between our dataset and the reference dataset. The x-axis denotes the statistical characteristics of the reference dataset, while the y-axis corresponds to those of the proposed dataset. Corr represents the pearson correlation coefficient. *p* stands for probability value.

Code availability

In this study, data processing, analysis, and visualization were carried out using Python 3.9 within Jupyter Notebooks. The complete codebase is available on GitHub: https://github.com/Ariza-Sun/MetroFlow. The origin-destination data and in-out flow data are too large to be hosted on a GitHub repository but can be found at figshare[26]. Apart from these two files, all other data and metadata have been uploaded to the repository for reference.

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Author contributions

P.S. and J.Y.: conceptualization, data curation, and writing original draft. Z.H., S.H. and S.X.: data validation. W.W., W.G. and Y.F.: formal analysis and visualization. X.Z. and T.Y.: data collection. X.Y., Y.J. and Y.X.: supervision and direction. P.S. and J.Y. contributed equally to this work. All authors participated in the review and editing of the manuscript.

Competing interests

The authors declare no competing interests.

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